

Indian Air Quality Index Analysis

End-to-End Data Analysis and Visual Storytelling

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Project Overview

This project analyzes Indian Air Quality data collected from various monitoring stations across India.

The goal of the project is to:

- Clean and preprocess the data
- Explore pollution patterns
- Perform statistical analysis
- Create visualizations
- Build an interactive Streamlit dashboard

Dataset Information

Source:

- Kaggle
- data.gov.in

Dataset contains:

- State
- City
- Station
- Pollutant Type
- Pollution Values
- Date and Time
- Latitude
- Longitude

Total Records: 4571

Total Columns: 11

Data Cleaning Summary

The following steps were performed:

- Checked missing values
- Removed duplicate records
- Converted date columns
- Created new features

New Features:

1. Pollution Range
2. Update Hour
3. Pollution Category

Cleaned dataset was saved as:

cleaned_air_quality.csv

Exploratory Data Analysis

Key Findings:

- PM2.5 was one of the most frequently monitored pollutants.
- Some states recorded significantly higher pollution levels.
- Moderate and Poor categories were common.
- Pollution varied across cities and pollutants.

Statistical Analysis

Three hypothesis tests were performed.

Test 1: PM2.5 vs PM10

Result: Insert your p-value and conclusion.

Test 2: North India vs South India

Result: Insert your p-value and conclusion.

Test 3: Pollution Category vs Pollutant Type

Result: Insert your p-value and conclusion.

Correlation analysis showed relationships among pollution indicators.

Visualization Insights

The project contains:

- 8 visualizations
- 1 animated Plotly chart

Major Insights:

- Pollution varies significantly across states.
- PM2.5 contributes heavily to pollution.
- Pollution indicators are strongly correlated.
- Pollution changes over time and location.

Streamlit Dashboard

Dashboard Features:

- KPI Cards
- Sidebar Filters
- Interactive Charts
- Correlation Heatmap
- Animated Plotly Visualization
- Dataset Viewer

The dashboard allows users to interactively explore air quality data.

Conclusion

This project successfully demonstrated an end-to-end data analytics workflow.

Completed Tasks:

- Data Cleaning
- EDA
- Statistical Analysis
- Visualization
- Dashboard Development

The project provides useful insights into air quality trends across India.